Retail Inventory Analytics and Supply Chain Dashboard using Microsoft Excel

Project Overview Report

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Tool Used

Microsoft Excel

Project Domain

Inventory Analytics and Supply Chain Management

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1. Introduction

Effective inventory management is essential for ensuring product availability, reducing operational costs, and supporting efficient supply chain operations. Organizations rely on inventory analytics to monitor stock levels, identify high-priority products, evaluate supplier performance, and forecast future demand.

This project applies inventory analytics techniques using Microsoft Excel to transform raw inventory data into actionable business insights. Through the use of data visualization, inventory classification, demand forecasting, and dashboard reporting, the project demonstrates how analytical methods can support inventory planning and operational decision-making.

2. Project Objectives

The primary objectives of this project were:

- To analyse inventory distribution across multiple product categories
- To evaluate supplier contribution to inventory availability
- To identify fast-moving products using inventory turnover analysis
- To classify inventory items using ABC Analysis
- To analyse monthly demand trends for selected products
- To forecast future demand using a 3-Month Moving Average model
- To develop an interactive executive dashboard for inventory monitoring and reporting

3. Dataset Overview

The dataset consists of retail inventory records across multiple product categories and suppliers.

Key fields analysed include:

- Item ID
- Item Name
- Category
- Supplier
- Stock Quantity
- Units Sold
- Inventory Turnover Ratio
- Stock Status
- Monthly Demand Data

The inventory dataset contains 40 items across categories such as Grocery, Dairy, Beverages, Fruits, Vegetables, Household, Personal Care, Bakery, Frozen Foods, Snacks, and Packaged Food.

4. Methodology

The project followed a structured inventory analytics workflow.

First, raw inventory data was organized and prepared for analysis. Supporting calculations were performed to derive inventory turnover ratios, stock status indicators, and ABC classifications.

Next, Pivot Tables and Pivot Charts were used to analyse inventory distribution, supplier contribution, and product movement patterns. Monthly demand trends were evaluated for selected high-priority products.

A 3-Month Moving Average forecasting model was then applied to estimate future demand based on historical demand patterns.

Finally, an interactive dashboard was developed to consolidate key performance indicators and provide a comprehensive view of inventory performance.

5. Tools and Analytical Techniques Used

The project was developed using Microsoft Excel and incorporates the following analytical techniques:

- Pivot Tables
- Pivot Charts
- Conditional Formatting
- KPI Reporting
- ABC Inventory Classification
- Inventory Turnover Analysis
- Monthly Demand Analysis
- Demand Forecasting
- Interactive Slicers
- Dashboard Development
- Data Visualization

6. Project Components

The completed project consists of the following analytical modules:

Executive Dashboard

Provides a consolidated view of inventory performance through KPIs, charts, and interactive filters.

ABC Analysis

Classifies inventory items into A, B, and C categories based on their relative importance and inventory movement.

Category Analysis

Evaluates inventory distribution across product categories to identify major inventory contributors.

Supplier Analysis

Analyses supplier contribution and procurement dependence across the inventory portfolio.

Inventory Turnover Analysis

Identifies fast-moving inventory items based on turnover performance.

Monthly Demand Analysis

Examines historical demand patterns for selected inventory items.

Forecast Analysis

Uses a 3-Month Moving Average model to estimate future demand trends.

7. Business Value

The analysis performed in this project can support inventory and supply chain decision-making by :

- Improving inventory visibility
- Supporting inventory prioritization
- Enhancing replenishment planning
- Monitoring supplier contribution
- Reducing stockout and overstock risks
- Supporting data-driven inventory management decisions
- Improving demand forecasting capabilities

8. Key Skills Demonstrated

This project demonstrates practical skills in:

- Inventory Analytics
- Supply Chain Analytics
- Data Analysis
- Business Analysis
- Dashboard Development
- KPI Reporting
- Demand Forecasting
- Data Visualization

- Microsoft Excel
- Pivot Tables and Pivot Charts
- Inventory Performance Monitoring

9. Project Deliverables

The final project includes:

- Project Overview Sheet
- Executive Dashboard
- ABC Analysis
- Category Analysis
- Supplier Analysis
- Inventory Turnover Analysis
- Monthly Demand Analysis
- Forecast Analysis
- Raw Data and Calculations
- Inventory Insights Report

10. Conclusion

This project demonstrates the practical application of inventory analytics techniques using Microsoft Excel. Through inventory classification, supplier contribution analysis, turnover analysis, demand forecasting, and dashboard reporting, the project provides valuable insights into inventory performance and supply chain operations.

The project highlights how data-driven analysis can support inventory planning, improve operational visibility, and assist organizations in making informed inventory management decisions. Additionally, it showcases analytical, reporting, and dashboard development skills that are relevant to inventory, operations, supply chain, and business analytics roles.